AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 21 · STUDENT LESSON

Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Name the traits. Add the gaps. Keep the ties.

Compare activity and book vectors under explicit rules without treating similarity as quality.

Keep one habit: Feature contract → coordinate gaps → tied book comparison → unchanged data with a changed weight.

MAIN LESSON · SECTION 1

Closer under which traits and distance rule?

> Keep feature meaning, order, scale and ties visible.

Your learning goals

I can represent items with numeric features.

I can calculate a simple **distance**.

I can explain how units and chosen features affect **similarity**.

Recall the size of an absolute gap

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What does abs(3 - 7) measure? If unfamiliar, calculate the positive gap.

Your end goal

By the end, I can calculate a feature **distance** and state how scaling changes its meaning.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 21: **Similarity** by Selected Traits

Different feature choices produce different notions of closeness.
</details>



THINKER MISSION

Compute simple **distance** on genre, length, and difficulty.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Define the two-feature activity representation

> This is a chosen comparison rule, not learned quality.

Task, feature order, scale and target

A fictional activity chooser represents each activity with energy and group-work ratings on a 0 to 5 **scale**. A learner requests [2, 4]. We compare options by adding absolute differences.

This is a chosen rule for **similarity**; it is not a **learned model** or an objective measure of quality.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **similarity** by selected traits to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Add the two absolute coordinate gaps

> Smaller means closer only under the declared representation.

Predict and run the complete activity script

```
target = [2, 4]
options = {"Sketch": [1, 3], "Relay": [5, 4]}
for name in options:
    features = options[name]
    distance = (abs(features[0] - target[0])
                + abs(features[1] - target[1]))
    print(name, distance)
```

Trace Sketch and Relay coordinate by coordinate

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Sketch [1, 3]	1 + 1	Distance 2
Relay [5, 4]	3 + 0	Distance 3

Read each row from left to right and connect the evidence to the stated rule.

abs gives a nonnegative gap. Under this rule Sketch is closer. Feature order must match: first energy, then group work. Both **scale**s are 0 to 5, but that does not prove one unit has identical practical meaning in both features.

Keep the features in the same order: energy first, group work second. For target [2,4] and Sketch [1,3], the gaps are abs(1-2)=1 and abs(3-4)=1; the **distance** is 2. Relay [5,4] has gaps 3 and 0, so its **distance** is 3.

Smaller means closer under this particular rule. It does not mean better in every way, because the two recorded features leave out other needs.

Explain what a pictured step means

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

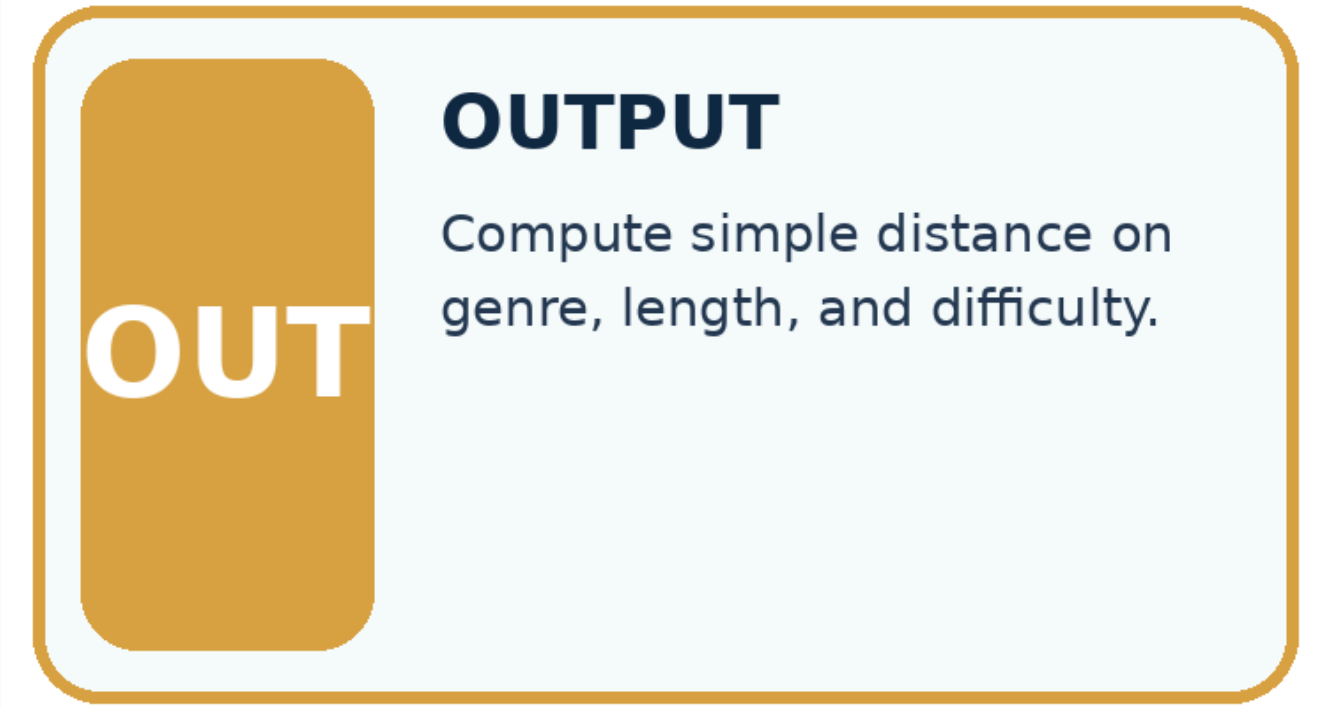
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Compute simple distance on genre, length, and difficulty.
↓
PROCESS
Different feature choices produce different notions of closeness.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

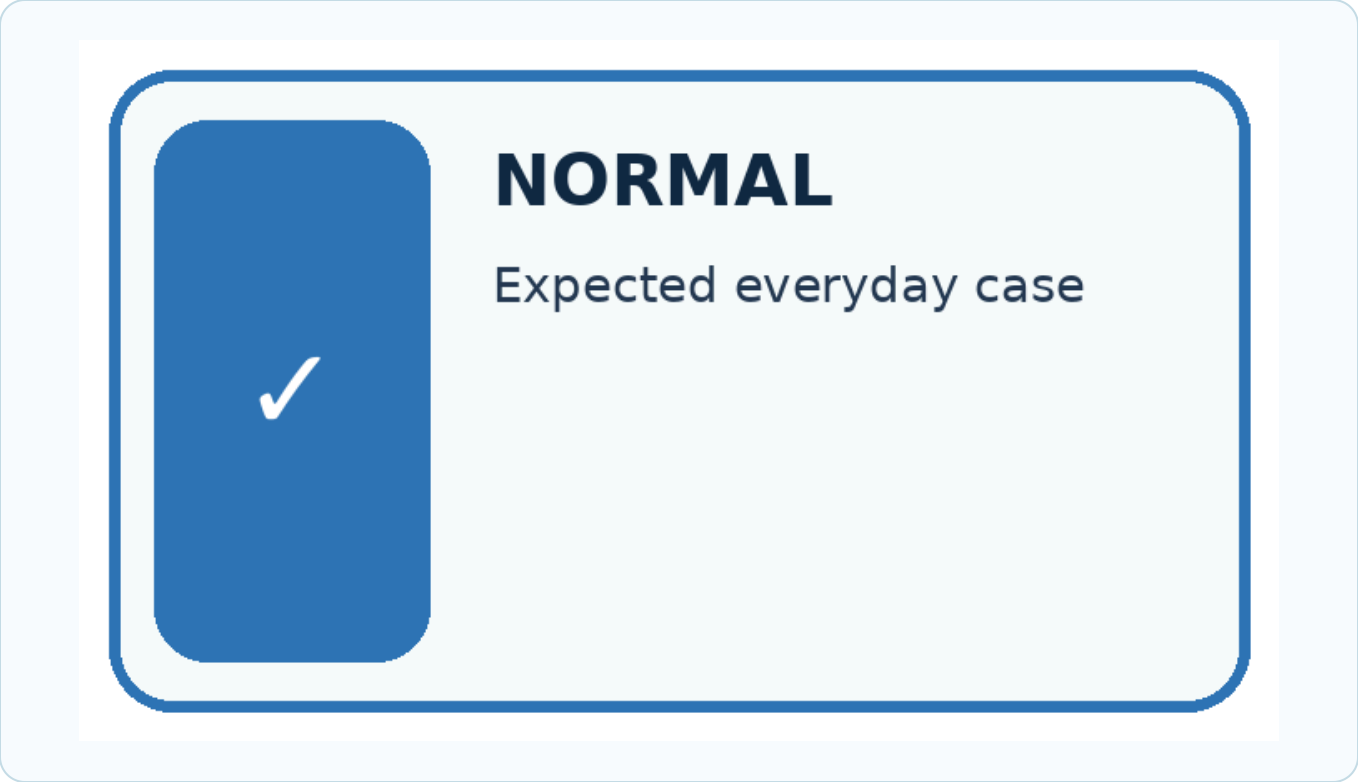
Calculate a gap one feature at a time

Connect the picture and the explanation

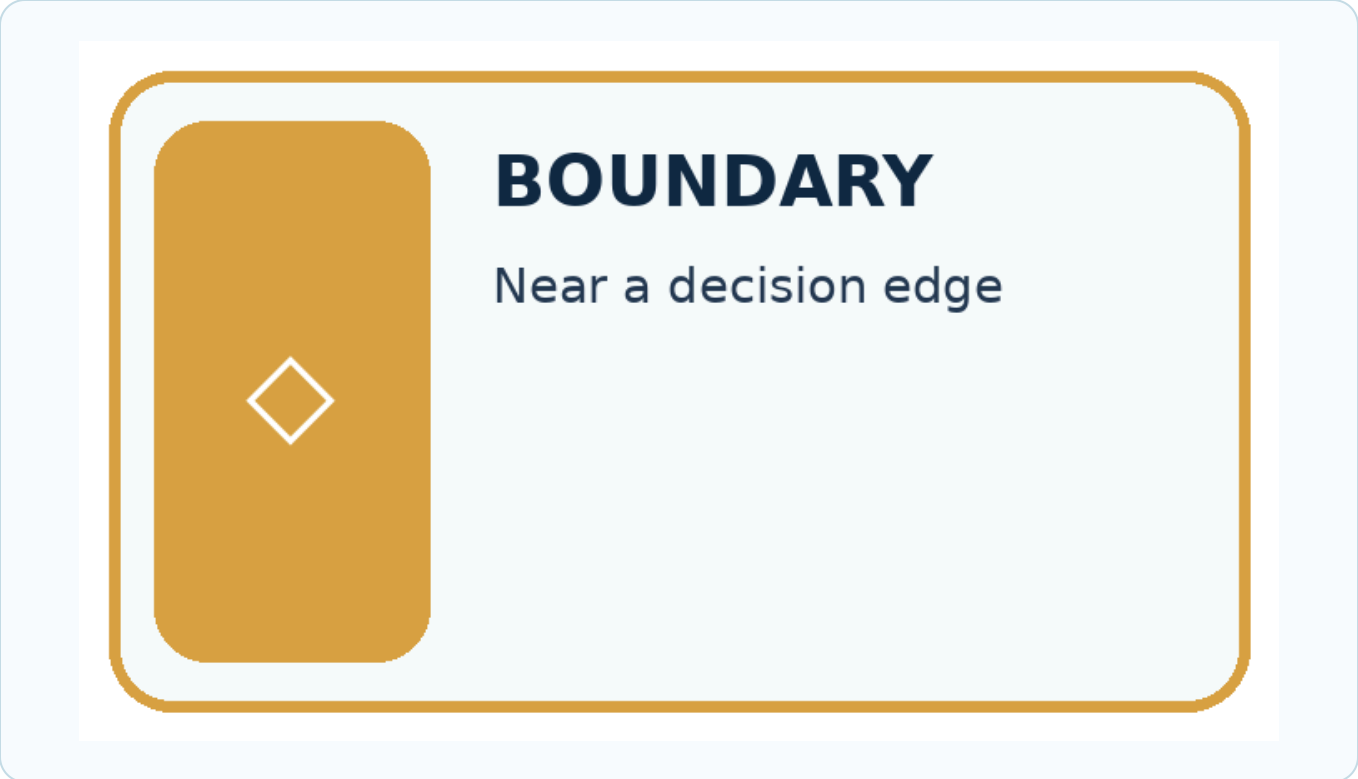
Compare ties and a changed feature weight

> Changing the comparison rule can change the answer without changing the data.

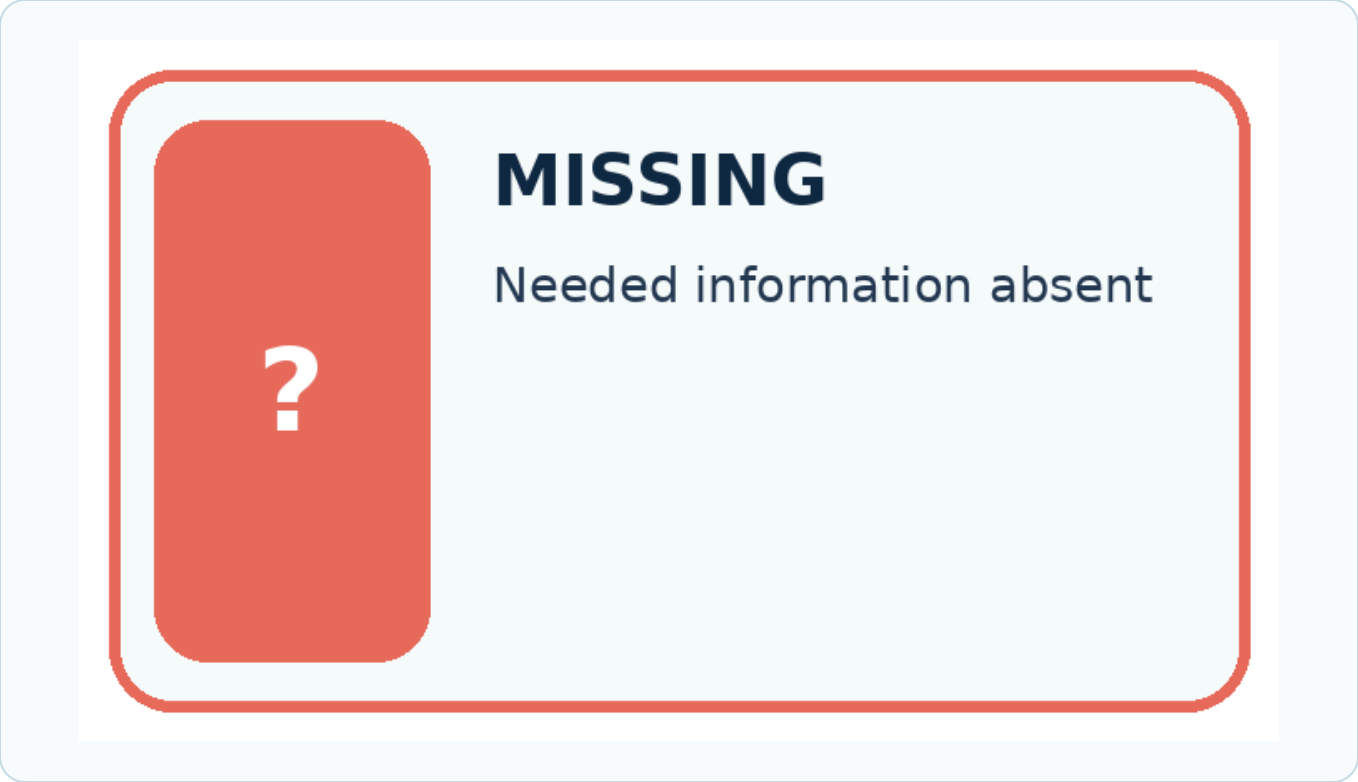
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Give the first-**coordinate** difference weight 2 while the second keeps weight 1.

▼ Check after predicting

Prediction: **Distance**s become A=3, B=4, C=3; A and C tie.

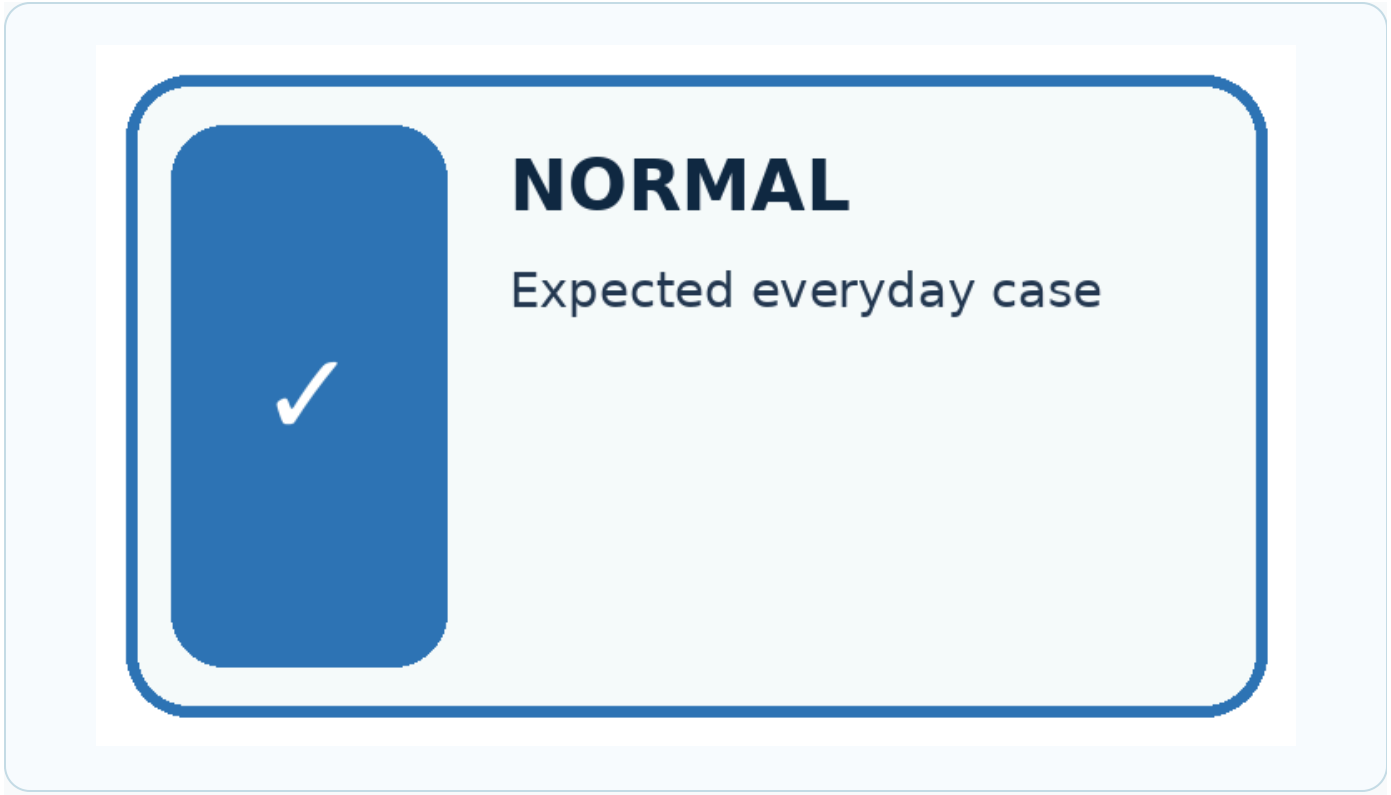
Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

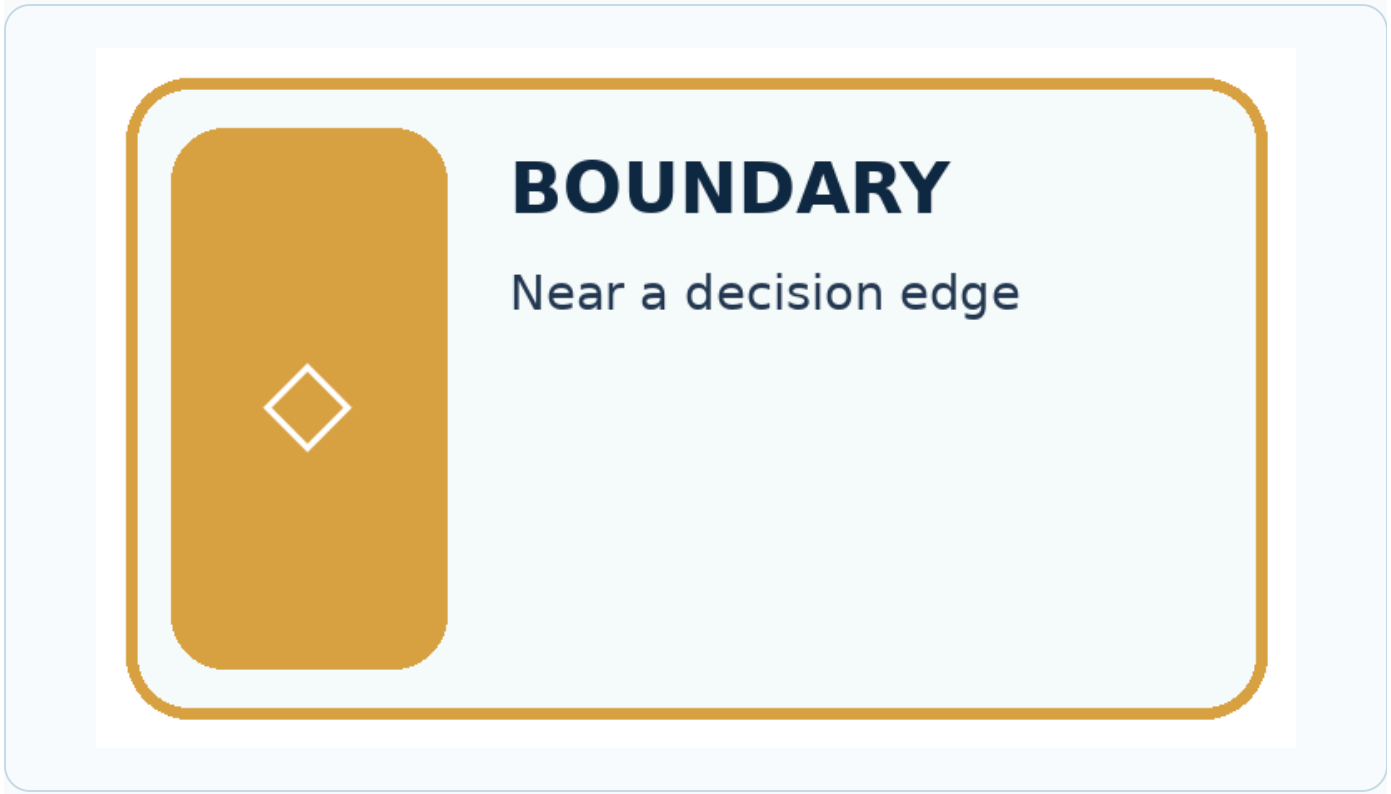
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

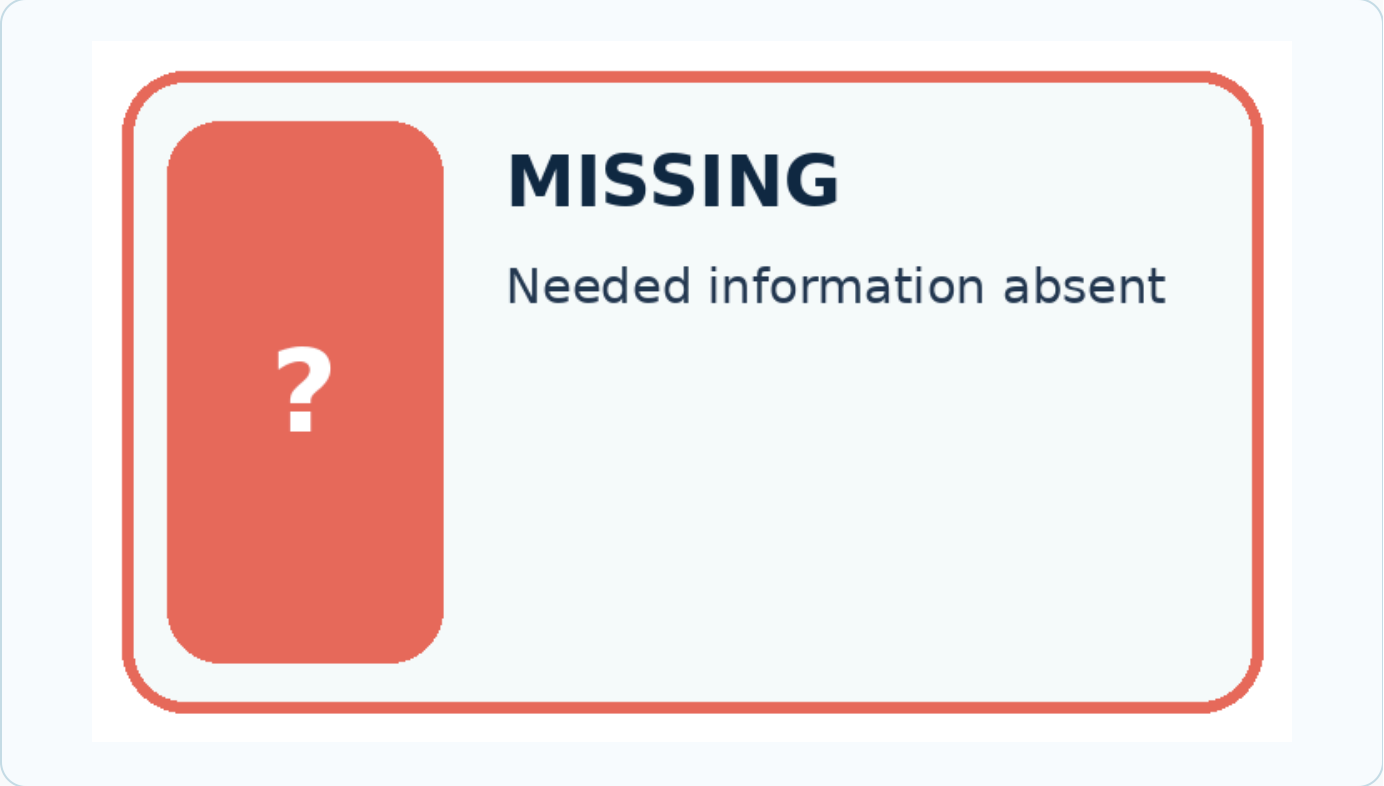
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace a separate book tie, then change its rule

> Keep the target and method beside every distance.

Practise two distances from the activity target

Find the distance from target [2, 4] to [2, 4] and to [3, 3].

Separate activity and book representations

The opening activity target is[2,4], with energy/group-work features. The separate book target below is[2,5]. For the book exercise, label the toy coordinates pace rating first and illustration rating second, each0–5. These invented ratings and their differences are declared teaching choices, not validated measurements of book quality. Keep the populations and feature meanings separate.

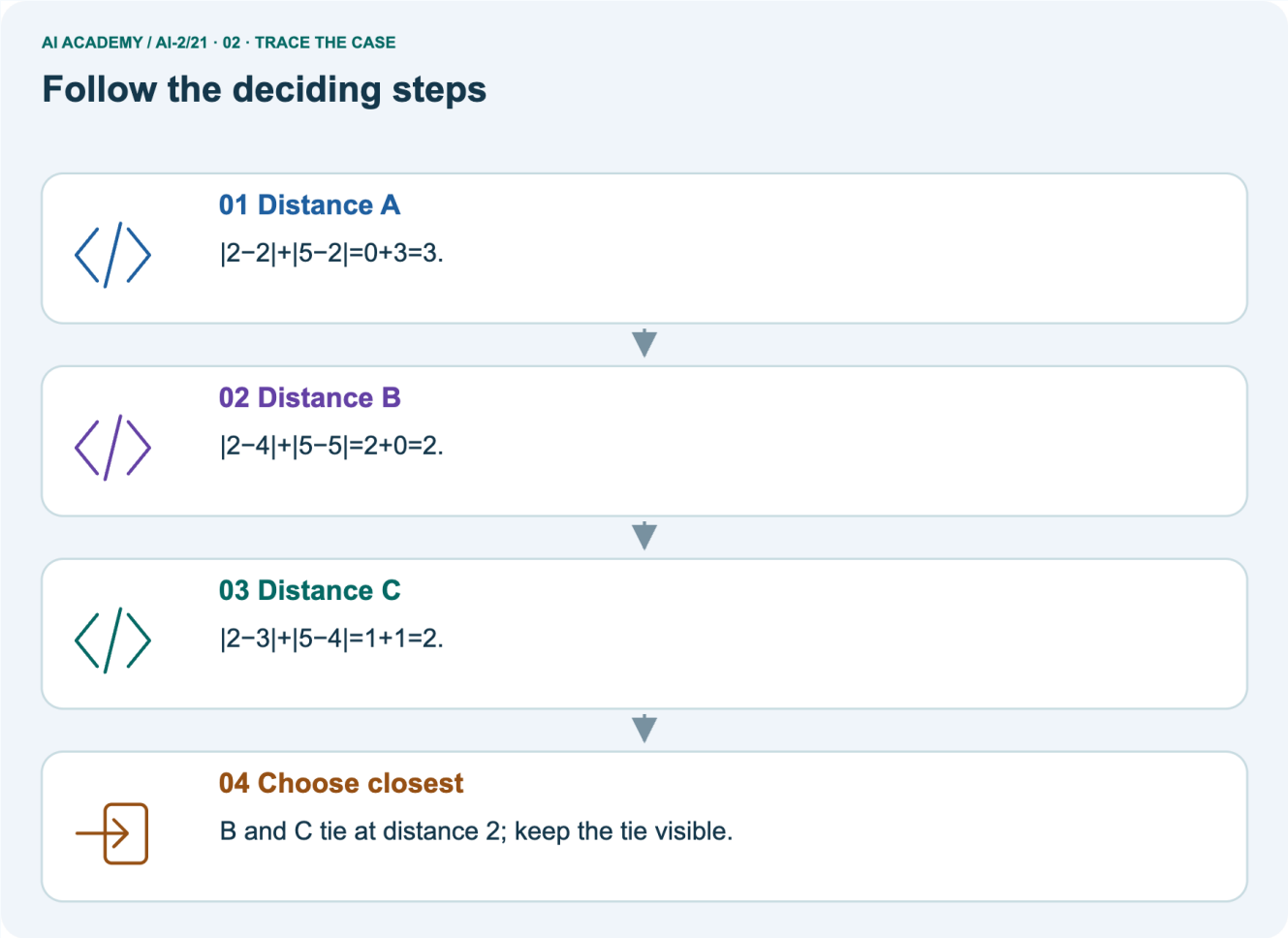
Read the book candidates and all-ties policy

A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Trace the original tie



Follow the labelled picture with the matching explanation. Open the vector diagram

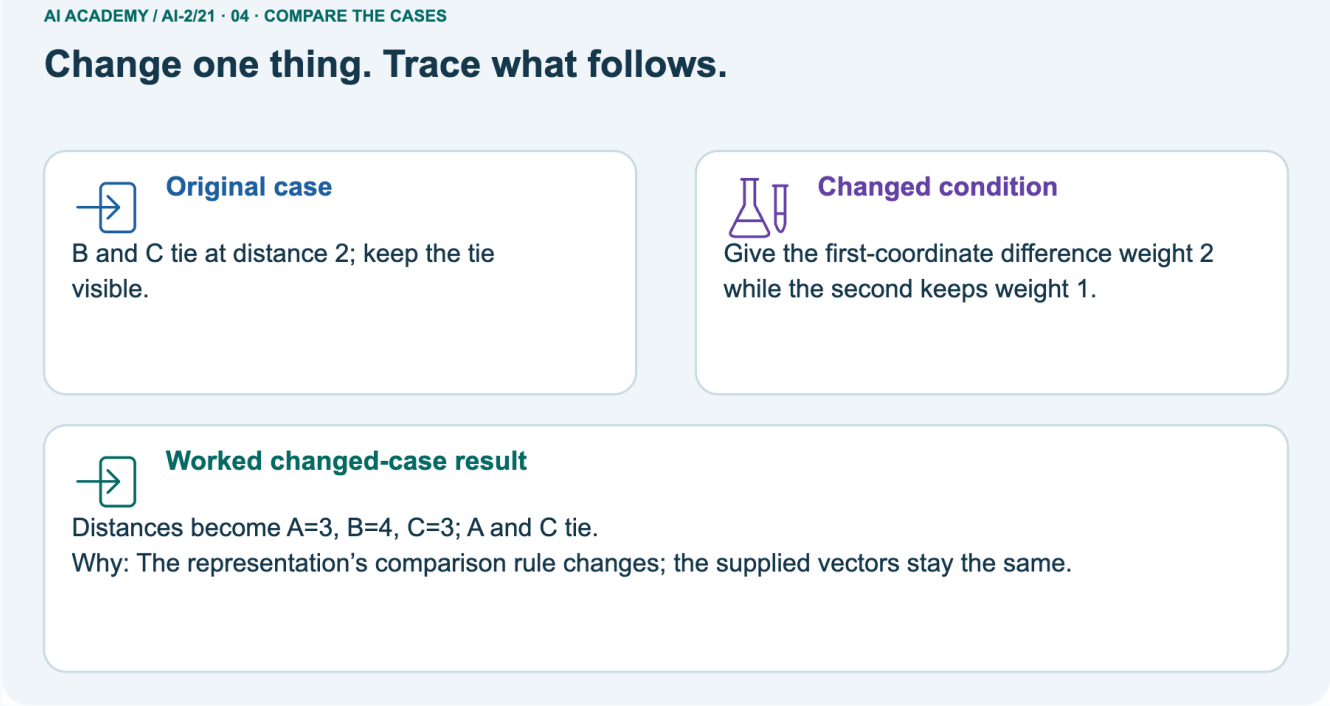
Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Predict first-coordinate weight2

- Change one thing: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain B’s changed status and the evidence limit

- Read the labelled picture: Explain why B ceases to be a closest option without any candidate data changing.
- A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 21 • WORKED EVIDENCE

1. Reader target: length=2, difficulty=3.
2. Book A: (2,2) → grid distance $|2-2|+|2-3|=1$.
3. Book B: (4,3) → distance $|4-2|+|3-3|=2$.
4. A is closer under these features; changing features can change the answer.

1. Reader target: length=2, difficulty=3.

2. Book A: (2,2) → grid **distance** $|2-2|+|2-3|=1$.

3. Book B: (4,3) → **distance** $|4-2|+|3-3|=2$.

4. A is closer under these features; changing features can change the answer.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Learn from the worked case

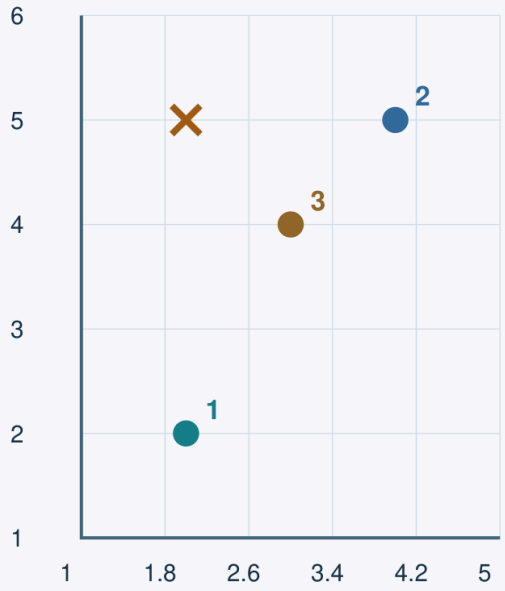
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Similarity by Selected Traits

The facts we will use

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance A

$|2-2|+|5-2|=0+3=3$.

2. Distance B

$|2-4|+|5-5|=2+0=2$.

3. Distance C

$|2-3|+|5-4|=1+1=2$.

4. Choose closest

B and C tie at distance 2; keep the tie visible.

CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Distance A	$\lvert 2-2\rvert+\lvert 5-2\rvert=0+3=3$.
2. Distance B	$\lvert 2-4\rvert+\lvert 5-5\rvert=2+0=2$.
3. Distance C	$\lvert 2-3\rvert+\lvert 5-4\rvert=1+1=2$.
4. Choose closest	B and C tie at distance 2; keep the tie visible.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

An exact feature match can still be unsuitable

> Check hard requirements separately from closeness.

State unrecorded-needs and representation limits

Similarity depends on the features collected and the **distance** rule. Missing accessibility, cost or preference information can make the closest item unsuitable. Hard requirements should be checked separately.

Challenge distance-zero implies equal suitability

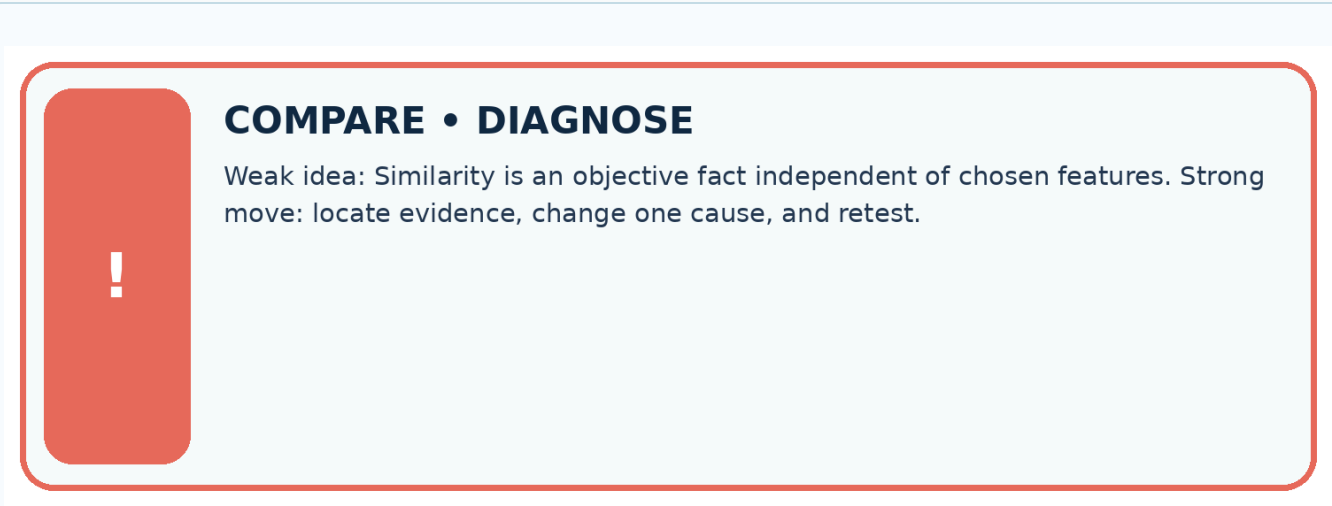
> **Two activities have distance zero. Must they suit every learner equally well?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		

Evidence record	Expected	Actual	Likely cause / next test
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Declare units, scaling and weights before ranking

> Do not change the scale silently to obtain a preferred winner.

Expose the unequal-scale problem

If one feature is duration in minutes up to 60 and another a rating up to 5, raw **distance** may be dominated by duration. Specify scaling or weights before ranking; do not silently change units to get a preferred winner.

Compute the standalone W2 distances

Use workbook W2 for the connected example "**Distance** between **feature vector**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge a successful-run guarantee

- Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the changed book-weight question

Explain why B ceases to be a closest option without any candidate data changing.

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Compute distances

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit six supplied vectors under two declared weight rules

> Show coordinates, each gap, totals and all ties.

The full six-book grid data

Toy book	Pace rating	Illustration rating
A	2	2
B	4	5
C	3	4
D	0	0
E	5	0
F	5	5

Predict, plot, compare and explain

1. Use all six supplied book vectors and target[2,5], with pace first and illustration second on toy0–5 scales. Plot horizontal pace and vertical illustration on a labelled grid. Predict each absolute gap, its equal-weight sum and all minimum-distance candidates. Then keep every vector fixed and change only the pace-gap weight to2; predict and recompute. Run the complete script or label a paper calculation. min finds the smallest distance; the final list comprehension retains every tied candidate. The program assumes complete numeric two-coordinate vectors; missing or mismatched fields need review before this calculation. Record what changed and do not claim that one rule is objectively best. Reserve the[3,2] independent comparison for W4.

Run the complete unchanged-data comparison

```
target = [2, 5]
books = {"A": [2, 2], "B": [4, 5], "C": [3, 4],
         "D": [0, 0], "E": [5, 0], "F": [5, 5]}
for weights in [(1, 1), (2, 1)]:
    distances = {}
    for name, features in books.items():
        distances[name] = (weights[0] * abs(features[0] - target[0])
                           + weights[1] * abs(features[1] - target[1]))
    smallest = min(distances.values())
    closest = [name for name in books if distances[name] == smallest]
    print(weights, distances, closest)
```

Keep the evidence and limit

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____

> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Plot six books on a 2-feature grid, compute simple grid distance, find nearest, then change one selected feature and compare.

Use the six supplied toy book vectors and target[2,5]. Label both grid axes, calculate equal-weight **distance**s, then change only the pace-gap weight to2. Keep all ties visible and distinguish changed rules from changed data.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: rank the fresh vectors and preserve the tie

> Write both coordinate gaps before the total.

The independent target and options

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

The declared rule and scope

Use unweighted Manhattan distance: sum both absolute differences, in the same two-coordinate order for target and every option. Smaller is closer under this rule; report ties rather than inventing a unique order. These are toy coordinates, not a claim about real reader

suitability.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Rank independently

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Make selected traits and their **scale**s visible in every recommendation explanation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Explain exactly what the recommender calls similar

> Feature choices express the method, not an objective quality judgement.

This week’s concrete contribution

For the fictional recommender, publish each feature’s meaning, units or rating scale, weights and tie policy. Check eligibility or accessibility requirements before treating closeness as a usable recommendation. A toy coordinate match does not establish interpersonal similarity, enjoyment or suitability. Keep missing fields explicit and preserve both versions if you compare weight choices.

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

Keep the checkpoint and human decision

THIS WEEK’S CHECKPOINT Make selected traits and their **scale**s visible in every recommendation explanation.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: connect a distance to its representation

> Name the features, order, scale, rule and missing information.

Explain without notes

- Close your notes. Explain this goal in your own words: Represent items with numeric features

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
similarity	How alike two items are according to selected features and a declared comparison method.	
distance	A specified measure of how far apart two representations are.	
coordinate	A value identifying position along an agreed axis or within a space.	
scale	The units and range used to represent or compare a quantity.	

Terms for the connected explanation

Word	Meaning
Feature vector	An ordered list of numeric features
Manhattan distance	The sum of absolute feature differences

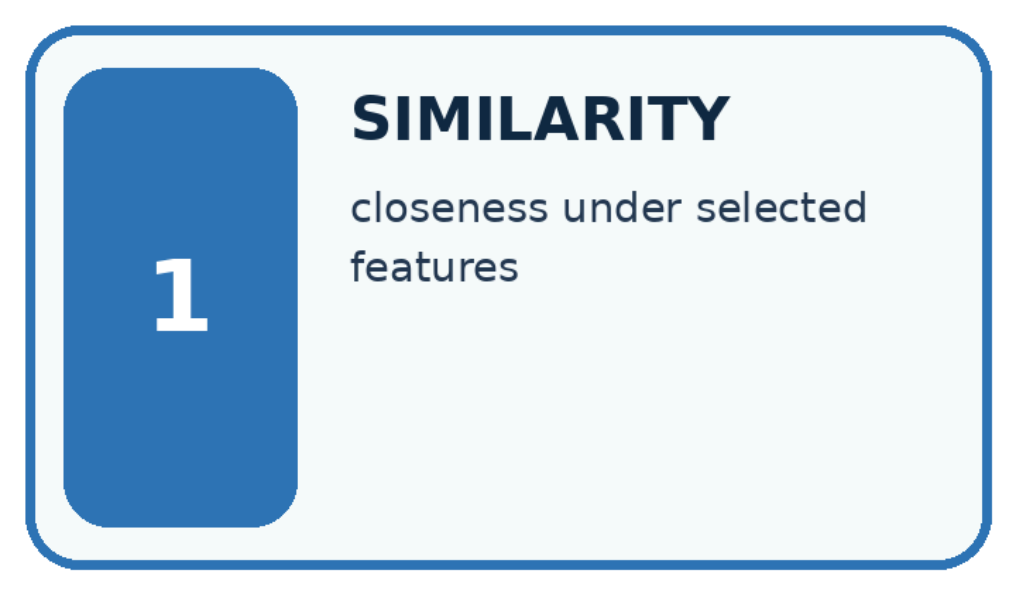
Word	Meaning
Normalization	Rescaling features to a comparable scale

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<summary>Earlier wording, alternatives and picture transcripts</summary>

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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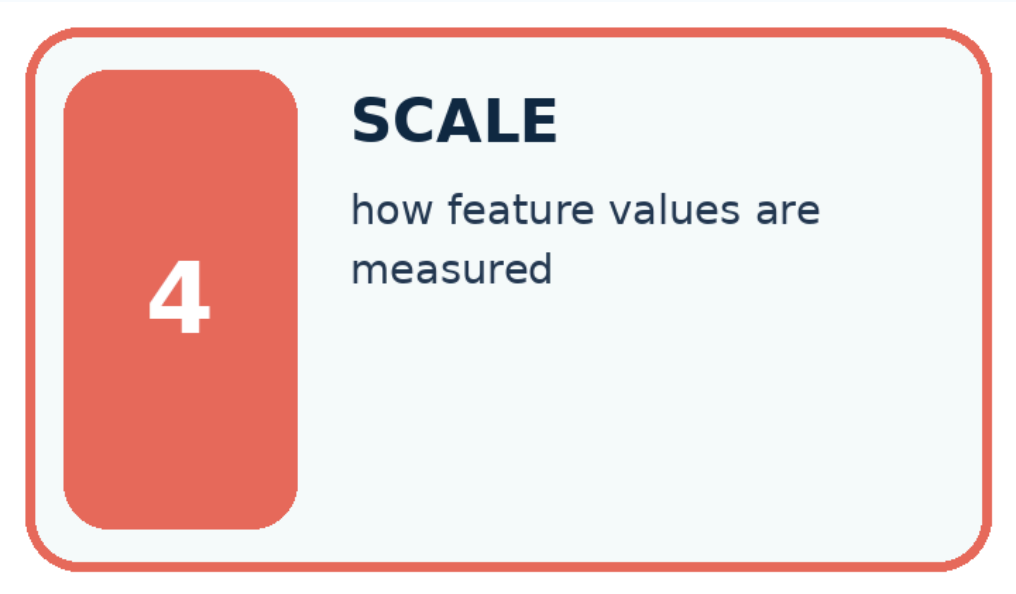
DISTANCE

number measuring difference

3

COORDINATE

feature value used as a position



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

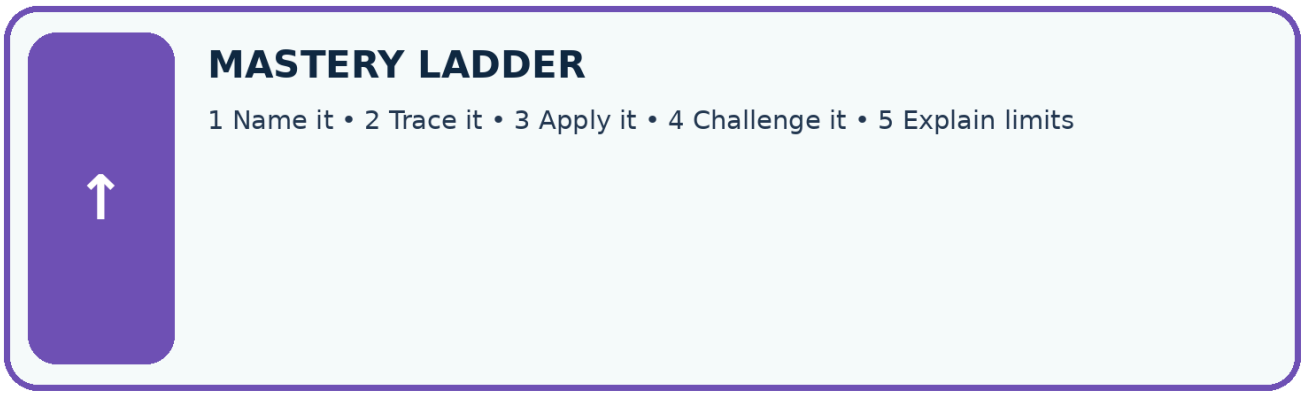
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Rank independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

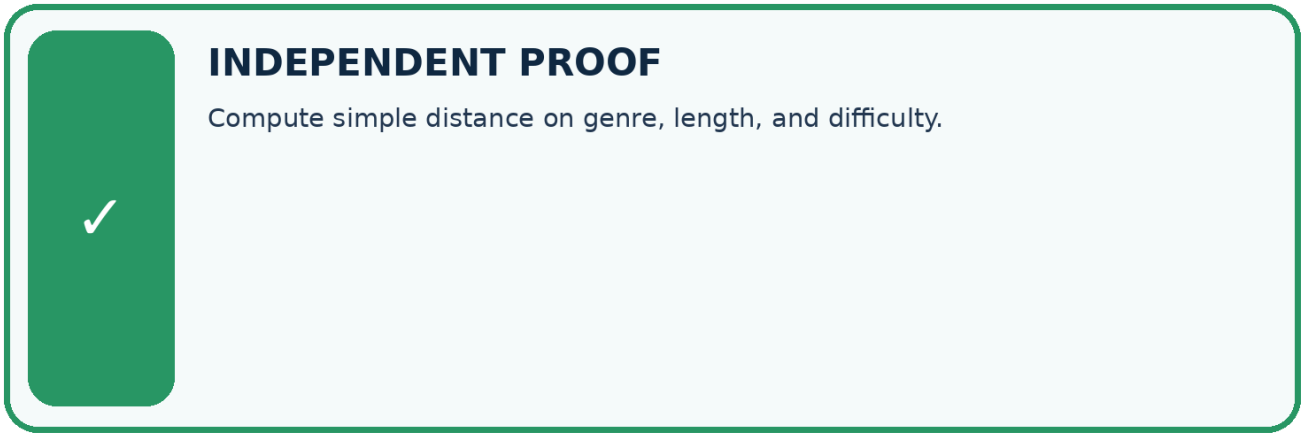
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Rank independently is the selected independent case.



EXPLANATION CHECK Keep feature meanings, scales and **distance** rules explicit, report all ties and limit similarity claims.

Write or draw your reasoning and evidence.

▼ **Extra practice: Count differing traits from query [1,0]**

REFERENCE · SECTION 13

Extra practice: Count differing traits from query [1,0]

> Try the question before opening its worked explanation.

Read the supplied case and question

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

Evidence or stage	Value or result
A [1,1]	1
B [1,0]	0
C [0,0]	1

▼ Earlier wording, alternatives and picture transcripts

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Count differing traits from query [1,0]

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

Distances: A=1, B=0, C=1. B is nearest with equal feature weights.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Represent items with numeric features	Explain the deciding step, show a trace or test, and state one limit.
Calculate a simple distance	Explain the deciding step, show a trace or test, and state one limit.
Explain how units and chosen features affect similarity	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional toy book matcher represents two preferences with numeric features.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy book matcher represents two preferences with numeric features.
- Which labelled step matches this detail? B and C tie at **distance** 2; keep the tie visible.
- 1. **Distance** C
- 2. Choose closest
- 3. **Distance** A
- 4. **Distance** B

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Choose closest B and C tie at **distance** 2; keep the tie visible. Each **coordinate** contributes according to the declared **distance** rule.

Smaller **distance** means closer in this representation, not automatically a better reading experience. If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important.

Feature units and scaling belong in the method record.

Changed-case question

- Changed-case practice: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?
- 1. **Distance**s become A=3, B=4, C=3; A and C tie.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair and save your work

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Similarity by Selected Traits: the case

AI ACADEMY / AI-2/21 · 01 · LOOK AT THE FACTS

Similarity by Selected Traits: the case

The situation

A fictional toy book matcher represents two preferences with numeric features.

Supplied evidence

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].
Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

By the end, I can calculate a feature distance and state how scaling changes its meaning.

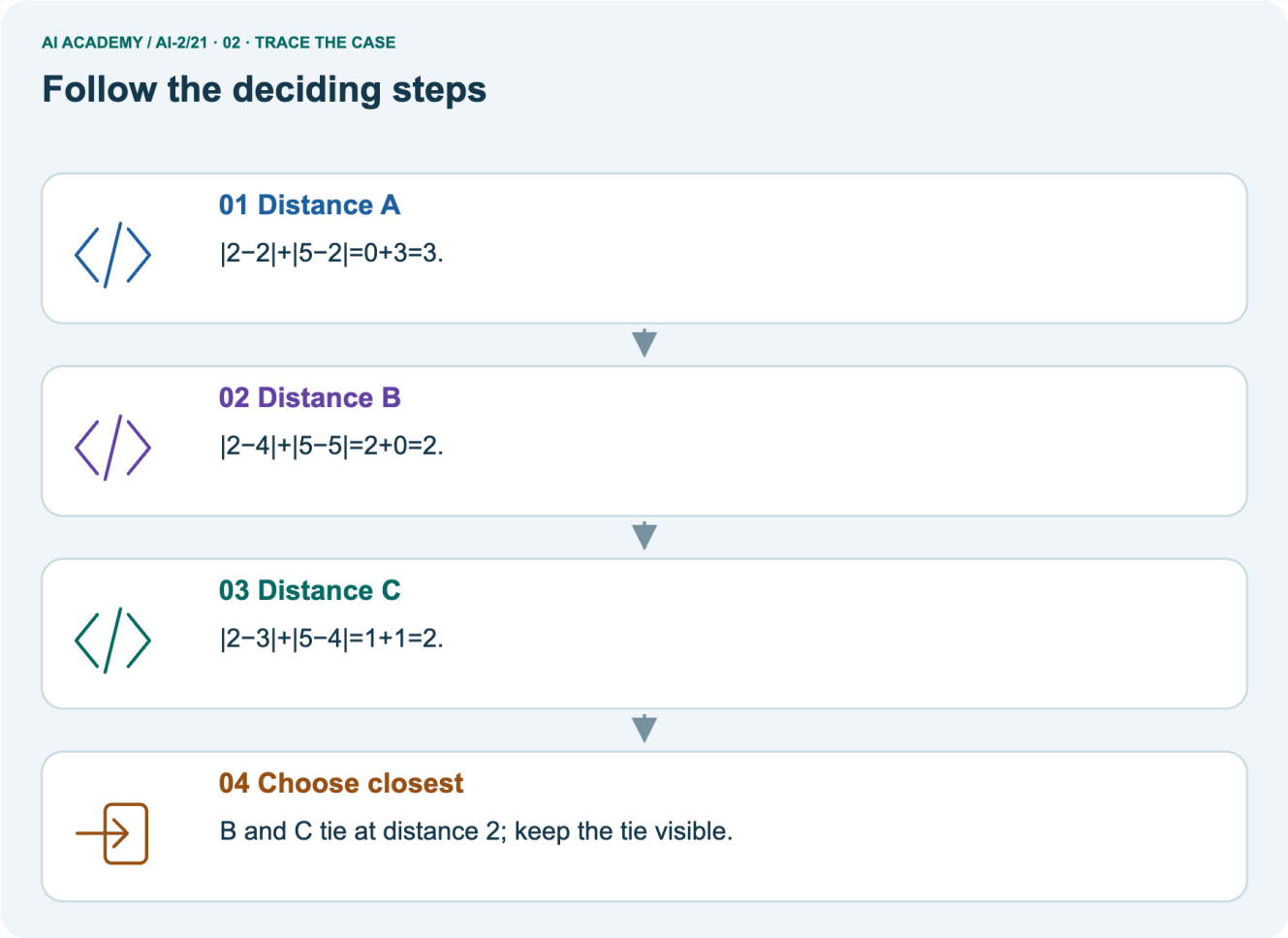
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

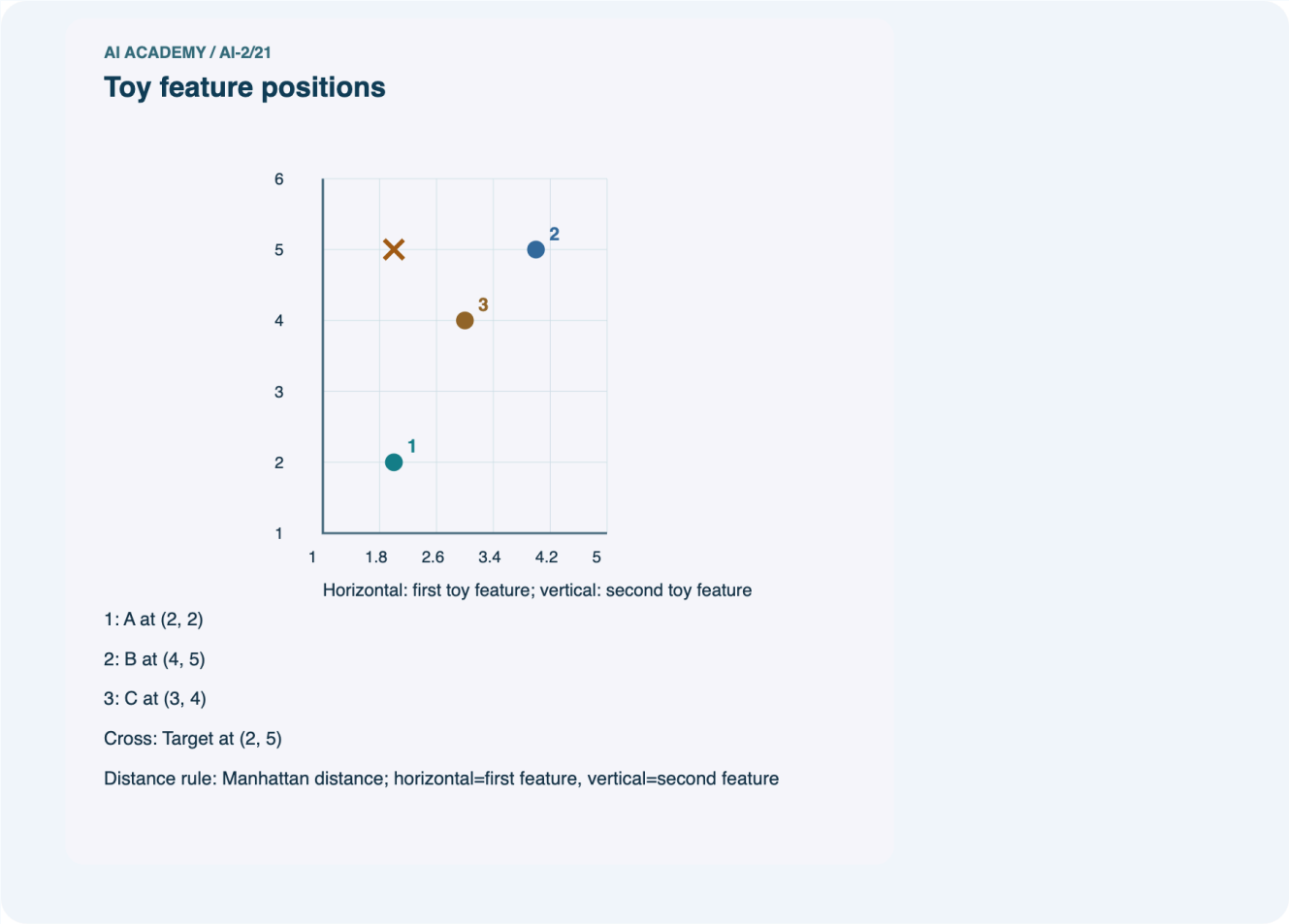


Follow the labelled picture with the matching explanation. Open the vector diagram

Distance A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

Point to each numbered stage and explain which recorded **input** it uses.

03. Toy feature positions

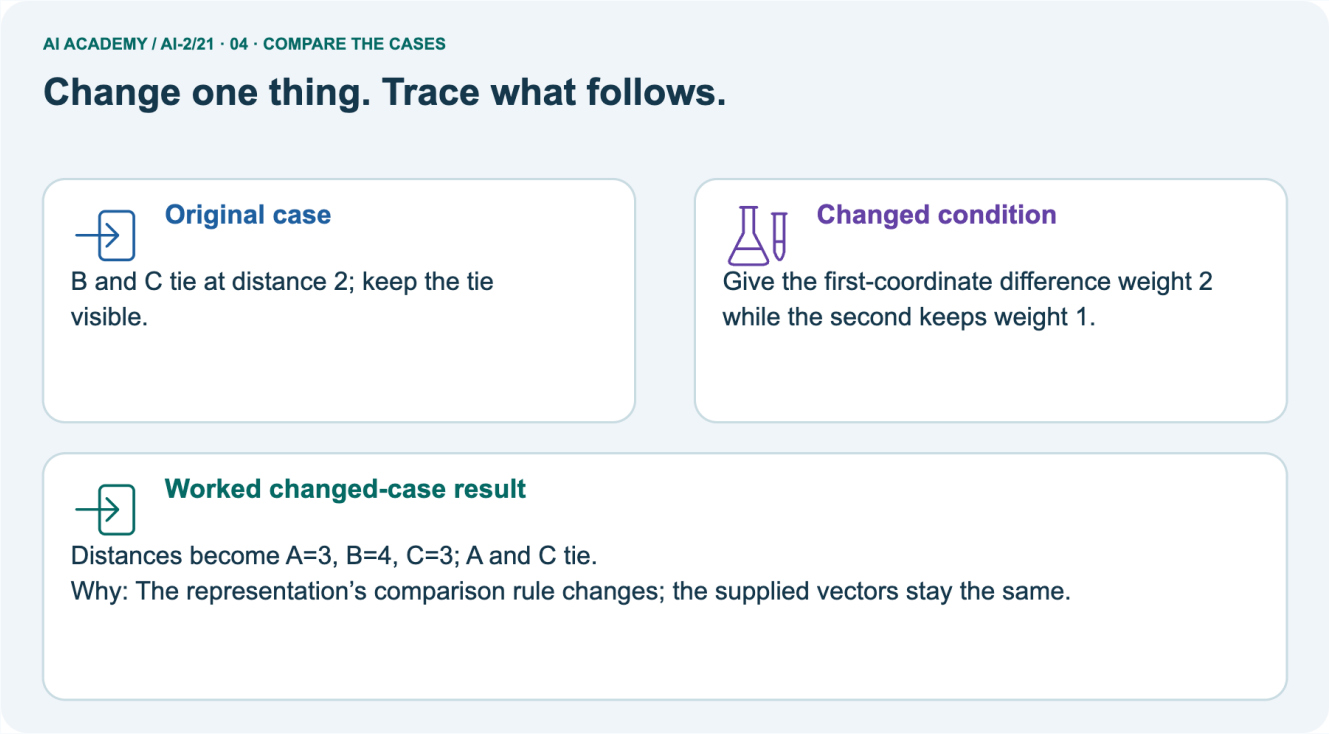


Follow the labelled picture with the matching explanation. Open the vector diagram

Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



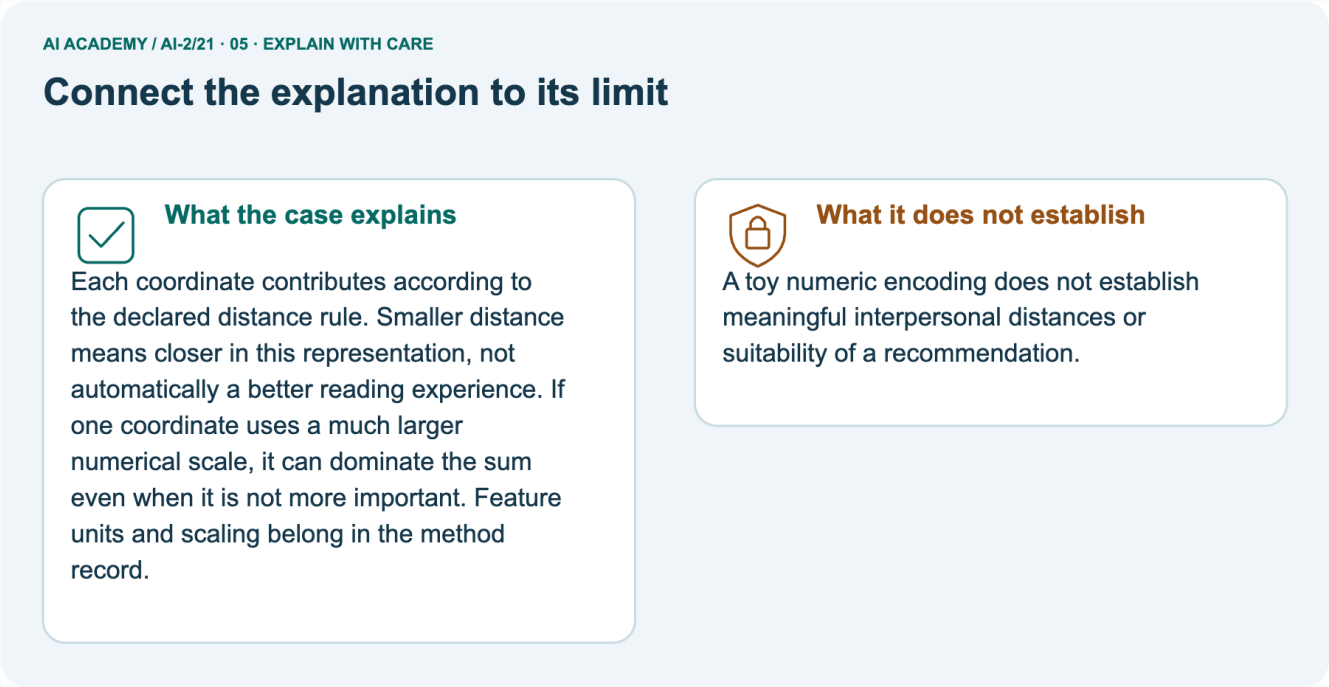
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation’s comparison rule changes; the supplied vectors stay the same.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

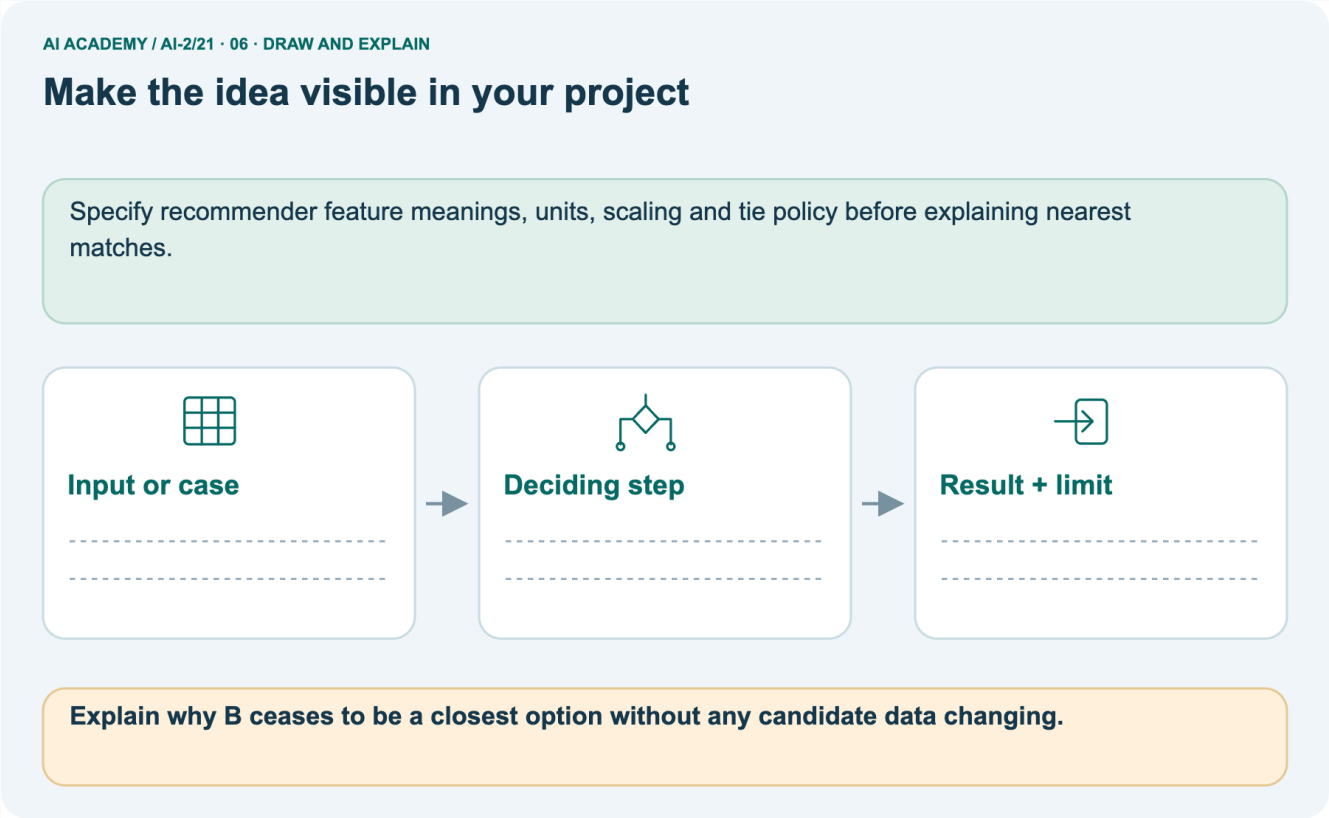
Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Similarity by Selected Traits: the case

02. Follow the deciding steps

03. Toy feature positions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.